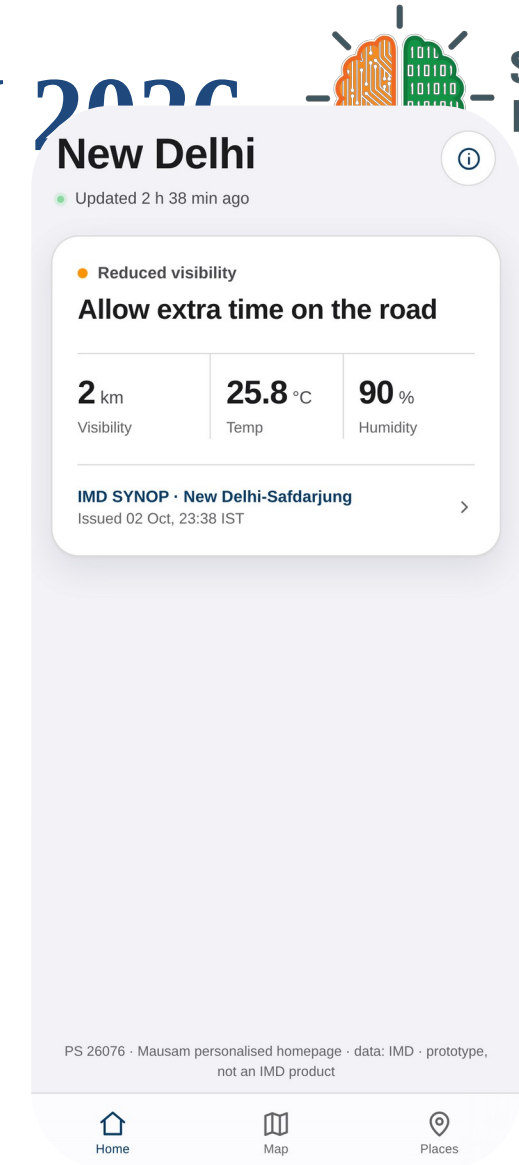


TITLE PAGE

- Problem Statement ID – SIH26076
- Personalised homepage for the Mausam mobile application
- Theme – Smart Automation
- PS Category – Software
- Team ID – 195697
- Team Name – Zencoderss

Mausam home: actions with proof

Every number shows its source, station, issue time and age. Every advisory can be re-derived by hand.



Working prototype home (illustrative screenshot).
Map + action cards with IMD provenance on every
number.

IDEA TITLE

Mausam home: actions with proof

PROPOSED SOLUTION (DESCRIBE YOUR IDEA/SOLUTION/PROTOTYPE)

A home screen that adapts to you: map on top, action cards below.

DETAILED EXPLANATION OF THE PROPOSED SOLUTION

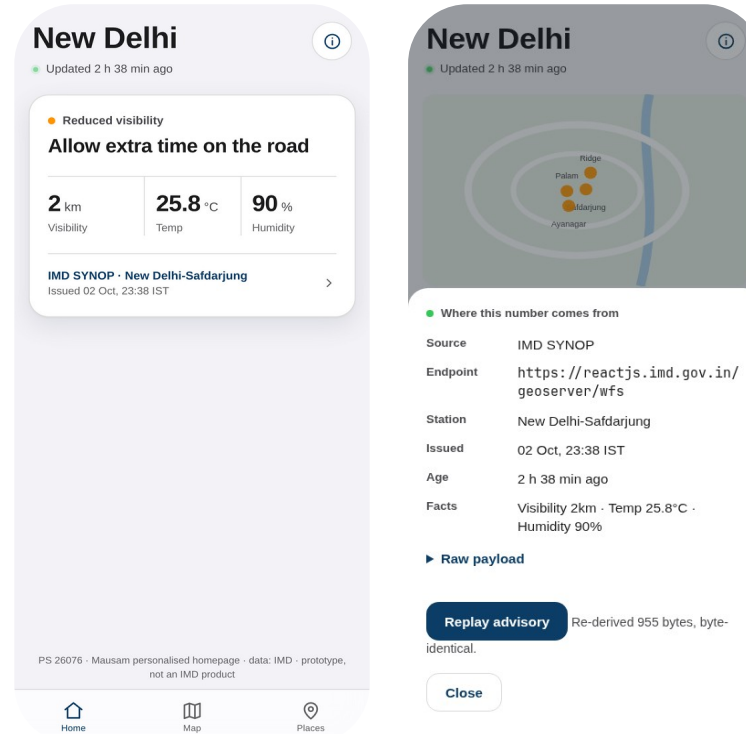
Onboarding picks your persona. The home reorders: the farmer sees sowing first, the commuter sees fog first.

HOW IT ADDRESSES THE PROBLEM

Every card says what to do, shows the readings behind it, and names its source, station and age.

INNOVATION AND UNIQUENESS OF THE SOLUTION

Source on every number. Replayable advice. Honest gaps where data is missing. A grounded bot that never invents numbers.



number on screen → tap ⓘ → source · station · issue time · age

Source on every number

station + issue time + age on each reading; tap ⓘ for the full provenance sheet

Advice you can replay

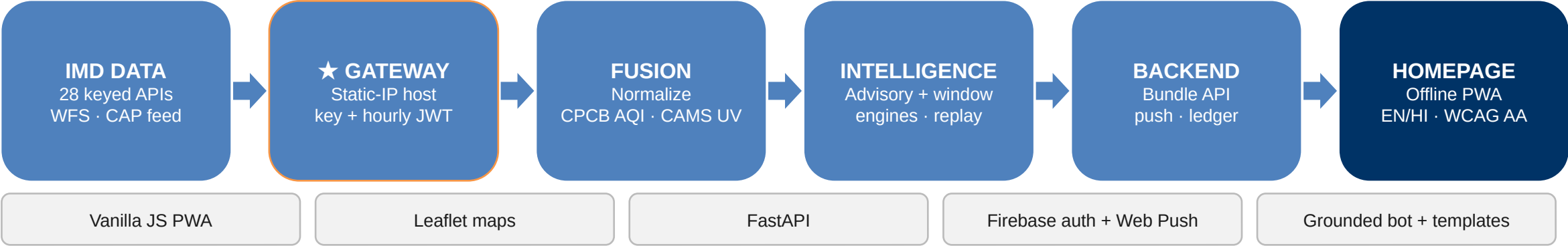
same data → same card, byte-identical

Honest gaps, no guesses

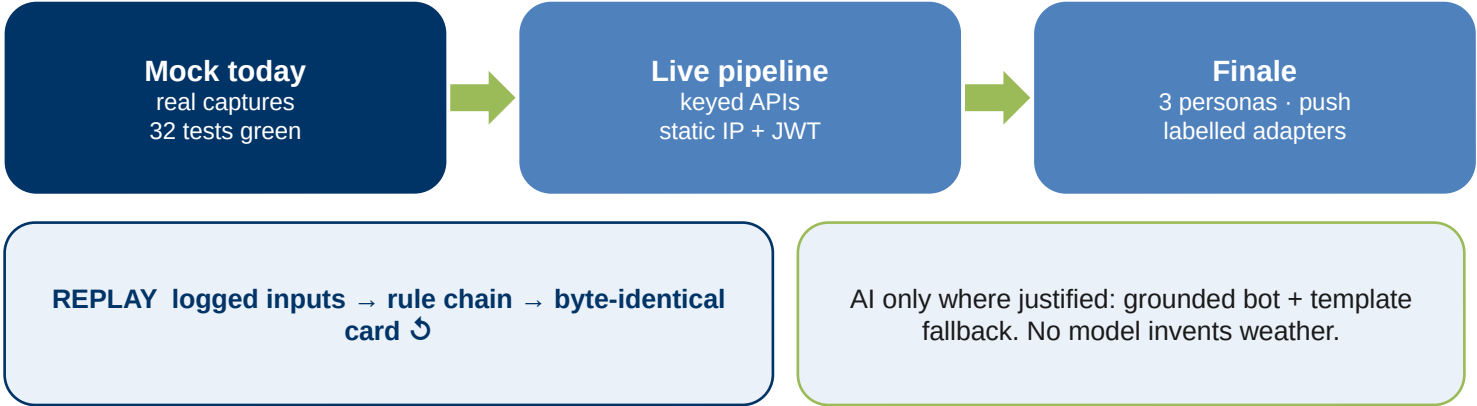
no data → a gap card, never a zero

Every number on this screen can tell you where it came from, when it was issued, and how old it is. Every advisory can be re-derived by hand.

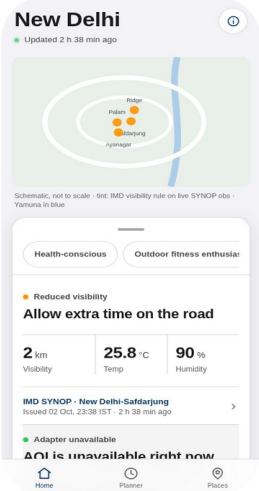
Technologies to be used (e.g. programming languages, frameworks, hardware)



Methodology and process for implementation (Flow Charts/Images/ working prototype)



Working prototype (illustrative screenshot)



replay reproduces this card byte-identically

Analysis of the feasibility of the idea

The mock runs offline on real captured IMD payloads. 32 tests green. Each risk below has a working fallback.

Potential challenges and risks

Keyed API needs a static-IP host

CPCB feed may be unreachable

Some IMD fields are undocumented

iOS limits push alerts

Strategies for overcoming these challenges

Run on saved captures + public WFS meanwhile

Labelled mirror fallback, else an honest gap card

Documented fallbacks; the screen states its data grain

Demo on Android; the inbox keeps every alert

Day 1 · static-IP host + key
registration

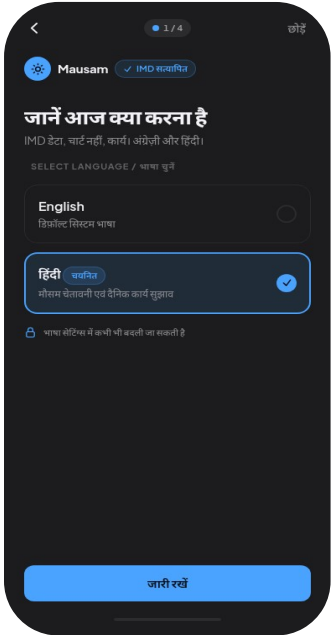
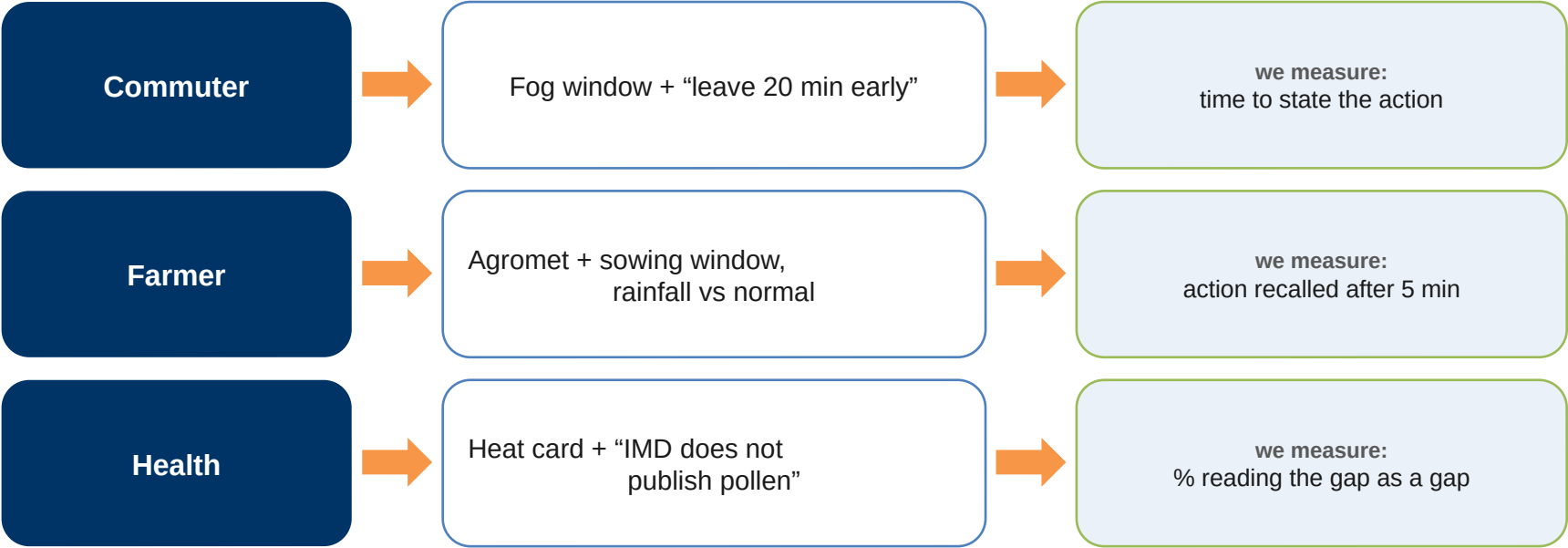
Week 1 · captured-payload mock
(done)

Week 3 · live keyed pipeline

Finale · demo: 3 personas push +
adapters

Viability: one container on a static-IP VM · one IMD API account (2 DEV + 2 PROD keys, attribution ledger on request) · no paid services · IMD data with attribution; CPCB/CAMS labelled adapters.

Potential impact on the target audience



Hindi onboarding +
offline shell (illustrative)

Benefits of the solution (social, economic, environmental, etc.)

Warning text is human-reviewed, never raw machine translation. No invented statistics: impact is measured in the pilot — can users state the action, how fast, does every card replay.

RESEARCH AND REFERENCES

IMD

official · keyed API

CPCB

official · labelled adapter

Copernicus

official · labelled

MeitY UX4G

design standard

Our mock

32 tests · replay

1. IMD API reference + portal user guide (key, hourly JWT, static IP) api.imd.gov.in
2. CPCB real-time AQI, GODL data.gov.in
3. Copernicus CAMS atmosphere data store ads.atmosphere.copernicus.eu
4. MeitY UX4G design system + PIB Mausam note (2023) ux4g.gov.in · pib.gov.in
5. Pai, N. et al. (2014), IMD Pune gridded rainfall 0.25° [published dataset](#)
6. Mausam app: existing Damini lightning + Meghdoot crop modules mausam.imd.gov.in
7. Working mock, fixtures, replay tests (32 pass) [submission folder](#) [Mausam/docs/evidence](#) (illustrative)